CROSSCODER MODEL DIFFING

FROM MODEL DIFFS TO CAUSAL CONTROL

A shared sparse representation for explaining and testing behavioral differences between code models

BEHAVIOR

↓

FEATURES

↓

STEERING

RESEARCH QUESTION

Evaluation tells us what changed. CrossCoder asks which internal differences can explain—and control—it.

BEHAVIOR

Which tasks changed?

pass → fail
fail → pass

What failure mode changed?

REPRESENTATION

Paired layer-16 residuals

Same-text alignment

Shared sparse features

CAUSAL TEST

Choose candidate directions

Intervene during generation

Compare against controls

Behavior localizes the model difference. Sparse features turn it into a testable hypothesis.

ONE ANALYSIS FRAMEWORK, TWO VERY DIFFERENT MODEL CHANGE

The same same-text CrossCoder design spans fine-tuning and model merging

DEEPSEEK

specialization through fine-tuning

BASE
layer 16
same evaluated token IDs

→

FINE-TUNED
layer 16
same evaluated token IDs

→

CROSSCODER
16,384 latents · ReLU
exact TopK-100

CODELLAMA

specialization through model merging

BASE
layer 16
same evaluated token IDs

→

MERGED
layer 16
same evaluated token IDs

→

CROSSCODER
16,384 latents · ReLU
exact TopK-100

Different specialization mechanisms; identical analytical question: which shared latent differences track—and control—behavior?

THE TWO SPECIALIZATIONS MOVE PERFORMANCE IN OPPOSITE DIRE

Canonical extraction-v4 labels · same raw generations, audited reprocessing

DEEPSEEK · BIGCODEBENCH

268/1140
BASE

404/1140
FINE-TUNED

+11.93 pp
BigCodeBench pass-rate change

CODELLAMA · BIGCODEBENCH

314/1140
BASE

27/1140
MERGED

-25.18 pp
BigCodeBench pass-rate change

DeepSeek: 215 improvements · 79 regressions

CodeLlama: 4 improvements · 291 regressions

RECATCHER CONTEXT

Directionally consistent motivation: specialization can improve aggregate capability while still creating localized regressions.

Not a direct numerical replication: ReCatcher used 10 generations/prompt and different regression definitions.

FEATURE SCREENING

We rank stable behavioral contrasts—not raw activation alone

4 SUMMARIES

max · early_max
mean · active fraction

→

Δ PER TASK

aggregate(specialized text)
− aggregate(base text)

→

OBSERVED EFFECT

mean Δ positives
− mean Δ controls

→

200 PERMUTATIONS

shuffle labels
recompute effect

→

E/V

effect ÷ null SD

Minimum support

max(3 tasks, 10% of positives)

Interpretation

E/V is permutation signal-to-noise—not a z-score or p-value.

Positive and negative orientations are retained: either enrichment direction may support behavior-aligned steering.

EIGHT COMPLEMENTARY SCREENING CELLS PER MODEL

Transition type × comparison design × temporal scope

DEEPSEEK

REGRESSION

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

IMPROVEMENT

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

CODELLAMA

REGRESSION

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

IMPROVEMENT

ONLY 4 POSITIVES · EXCLUDED

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

Local = ±10% around the first normalized divergence—not a semantic error category.

WHAT THE SCREENING ACTUALLY RETURNS

Representative valid cells; E/V is the primary ranking signal

DEEPSEEK · IMPROVEMENT ASSOCIATION

Rank	Feature	Effect	E/V
1	3602	-0.021	**-8.23**
2	10168	-0.018	**-7.67**
3	5422	-0.005	**-7.66**
4	1982	-0.015	**-7.53**
5	1862	-0.008	**-7.45**
6	1650	-0.026	**-7.28**
7	5039	-0.088	**-7.26**
8	9580	-0.013	**-7.24**
9	5837	-0.010	**-7.18**
10	9537	-0.038	**-7.15**

CODELLAMA · REGRESSION ASSOCIATION

Rank	Feature	Effect	E/V
1	13147	+0.126	**+5.24**
2	8992	-0.011	**-4.93**
3	1840	-0.046	**-4.77**
4	16161	+0.127	**+4.76**
5	15471	+0.005	**+4.75**
6	10570	+0.188	**+4.60**
7	8138	-4.973	**-4.51**
8	7353	+0.228	**+4.48**
9	14359	-0.166	**-4.46**
10	2825	+0.338	**+4.38**

Examples—not a single global ranking. Candidate pools are formed by union and deduplication across valid cells.

SCREEN FIRST; TEST CAUSALLY AT A COMMON OPERATING POINT

The statistical shortlist reduces the intervention search by ~99.8%

16,384 → **≈30–40** → **≈0.2%**

latents / CrossCoder

candidates / model

of latent space

WHY |α| = 3?

Earlier dose-response work suggested a practical exploratory point: visible causal changes below the most disruptive high-dose regime.

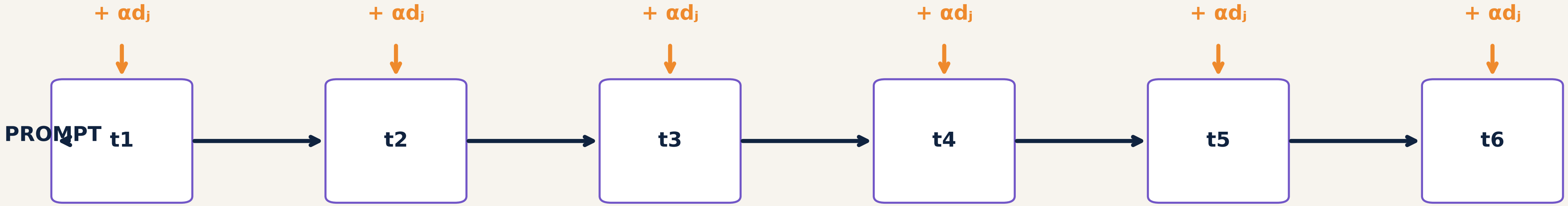
WHY STANDARDIZE?

One magnitude makes dozens of candidates comparable before full dose curves, reverse-model t

|α|=3 is an operating point—not an independently validated optimum.

OUR CURRENT INTERVENTION IS CONTINUOUS STEERING

The selected decoder direction is added at every autoregressive generation step



CURRENT PROTOCOL

Layer 16 · last residual position
feature decoder on intervened model side
applied through the full generation

WHAT IT IS NOT

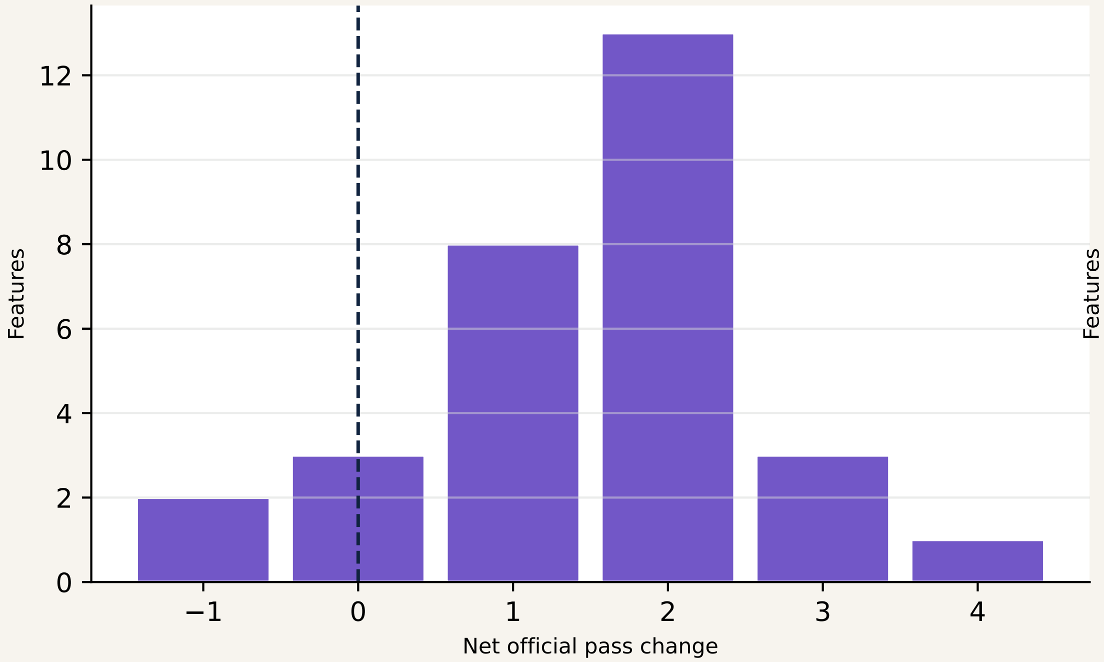
Not a one-token intervention
not activation-gated or clamped
not conditional on natural activation

This tests continuous control of the generation trajectory—not whether one naturally active token alone caused the error.

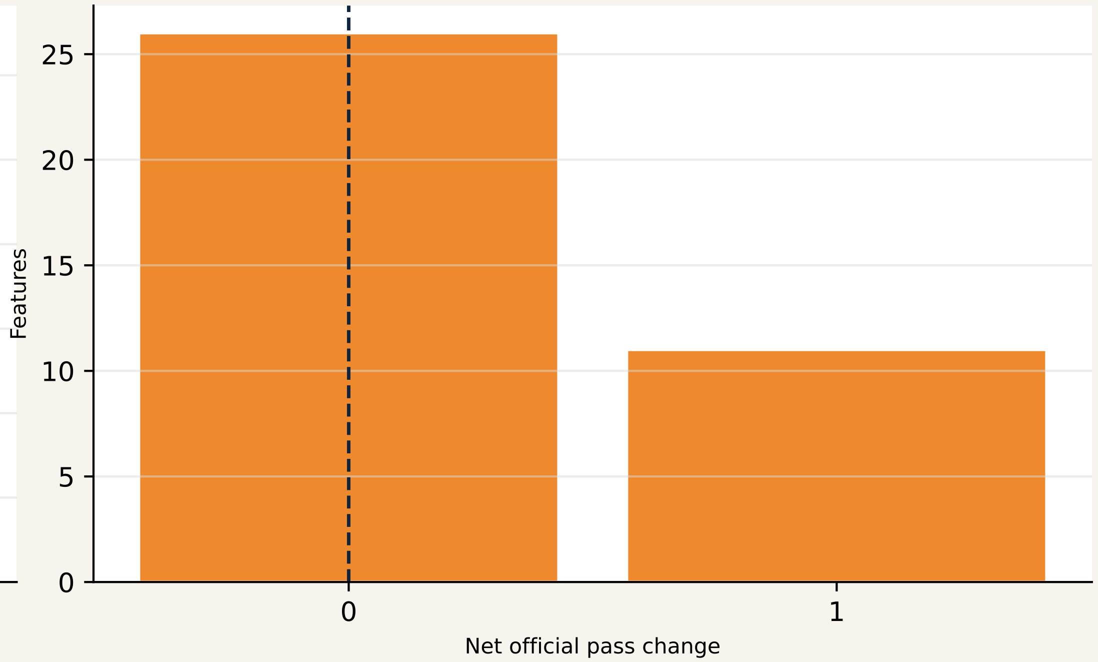
THE α=3 SWEEP REVEALS BOTH MODEL AND TASK SENSITIVITY

Official pass changes motivate—but do not replace—random and sham controls

DeepSeek · 30 features



CodeLlama · 37 features



TASK SENSITIVITY

DS /1030
29/30

CL /119
10/11

TARGET

screen-selected latent

RANDOM LATENTS

not selected by screen/α3

ORTHOGONAL SHAMS

norm-controlled directions

Controls are always good practice; concentrated successes make their necessity especially visible here.

TEN CANDIDATES ADVANCE TO BBASV

Causal outcome at |α|=3 first; applicable screening rank breaks ties

Model	Feature	Repair sign	Reverse model	Net Δ	Passes
DeepSeek	3048	−	FT	+4	19/80
DeepSeek	13801	−	FT	+3	18/80
DeepSeek	7828	−	FT	+3	18/80
DeepSeek	15669	−	FT	+3	18/80
DeepSeek	13191	−	FT	+2	17/80
CodeLlama	13147	−	base	+1	1/50
CodeLlama	12253	+	base	+1	1/50
CodeLlama	10570	−	base	+1	1/50
CodeLlama	14359	+	base	+1	1/50
CodeLlama	2310	+	base	+1	1/50

BBASV · DIRECT

Move the worse model toward the behavior-associated direction.

Full dose curve for each Top-5 feature.

BBASV · REVERSE + CONTROLS

Invert direction on the better model.

Compare random latents and orthogonal shams.

DeepSeek colors: purple · CodeLlama colors: orange

Bidirectional Behavior-Aligned Steering Validation